
2019-Ph-H1-Q9

Section: Particles and Waves

Topic: Refraction of Light

Question:

A ray of light passes from air into glass. Which quantity decreases?

- A) Frequency
- B) Wavelength
- C) Amplitude
- D) Angle of incidence
- E) Energy

Answer:

Wavelength decreases in a denser medium. Frequency stays constant.

Correct Option: B

Guidance for Students:

As light slows down, wavelength shortens — but frequency stays the same.

Revision Tips:

- **Speed = frequency × wavelength**
 - In denser medium: speed ↓, wavelength ↓, frequency = constant
 - This is why light **bends toward the normal**
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